# Connectivity Fault Management

**Revision history:**

| Revision | Author | Contact Info | Description | Date |
|---|---|---|---|---|
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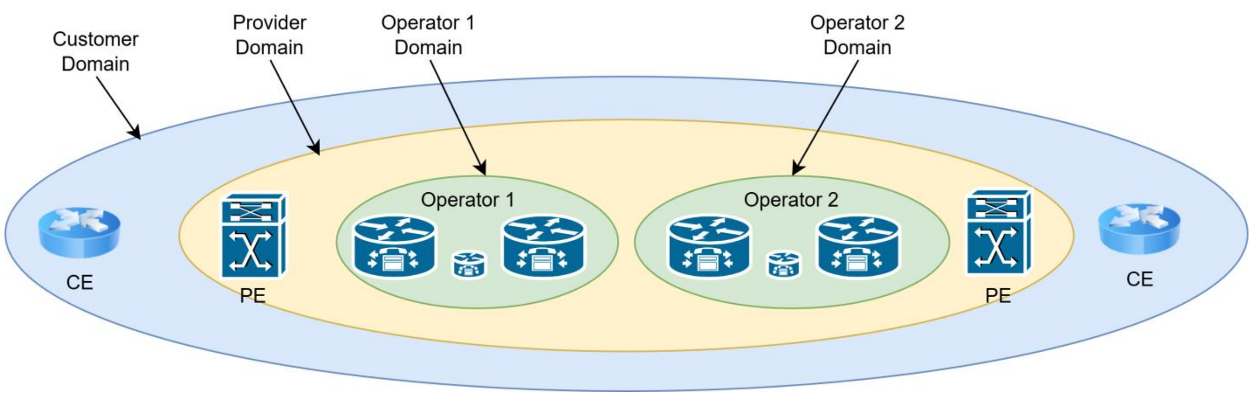
# 1. Introduction

Ethernet CFM, a network-level Ethernet OAM mechanism, supports functions including end-to-end connectivity fault detection, fault advertisement, fault acknowledgement, and fault locations. CFM monitors network connectivity and pinpoints connectivity faults. It is used with protection switching techniques to make networks more reliable.

CFM functions are partitioned as follows:
- ✓ Path discovery
- ✓ Fault detection
- ✓ Fault verification and isolation
- ✓ Fault notification
- ✓ Fault recovery

# 2. CFM Terminology




## 2.1 Maintenance domains

An MD is a section of the network managed by a single operator for monitoring and controlling connectivity. Each MD is assigned a unique level from 0 (smallest scope) to 7 (widest scope), ensuring that network areas do not intersect, though they can be nested or adjacent. In nested MDs, the outer domain can pass CFM messages transparently to inner domains for broader fault isolation. Key terms include:

- **MD Level:** Defines the hierarchy, where higher levels represent larger, often customer-facing domains (e.g., Customer MD Level 7, Operator MD Level 0).
- **MD Name:** Unique identifier for easy administration and responsibility tracking.

**Example**: In a customer-provider-operator scenario, Customer MDs may be assigned level 7, Provider MDs level 5, and Operator MDs level 0. This hierarchy ensures that each domain handles its own CFM messages without interference from others.

## 2.2 Maintenance Points (MPs)

Maintenance Points serve as reference points within an MD to handle and filter CFM frames, which identify and diagnose network connectivity issues. MPs can either terminate or pass CFM frames depending on their level and configuration. Types of MPs include:

- **Maintenance Endpoint (MEP):** Found at MD boundaries, MEPs initiate and terminate CFM messages.
- **Maintenance Intermediate Point (MIP):** Positioned within an MD to receive and relay CFM messages without initiating them.

**Example**: If a CFM message from a higher MD (e.g., level 7) arrives at a lower MD point (e.g., level 0), it will pass through without interference.

## 2.3 Maintenance Association (MA)

An MA is a group of MEPs within the same MD level, ensuring that a specific service instance (e.g., a VLAN) operates without faults. The MA operates at a specific MD level, and all CFM frames within the MA pertain to that level. Each MA has a unique identifier within the MD.

## 2.4 Maintenance Endpoint (MEP)

MEPs are located at the edge of an MD and MA. The service type and level of packets sent by a MEP are determined by the MD and MA to which the MEP belongs. A MEP processes packets at specific levels based on its own level, and sends packets carrying its own level. If a MEP receives a packet carrying a level higher than its own, the MEP does not process the packet and instead loops it along the reverse path. If a MEP receives a packet carrying a level lower than or equal to its own, it processes the packet.

MEPs are configured on interfaces. The MEP level is equal to the MD level.

MEPs configured on CFM-enabled devices are called local MEPs. MEPs configured on other devices in the same MA are called remote maintenance association end points (RMEPs).

MEPs are classified into the following types

- ✓ Inward-facing MEP: sends packets to other interfaces on the same device.
- ✓ Outward-facing MEP: sends packets to the interface on which the MEP is configured.

### 2.4.1 Up Mep

**Up MEPs** are designed to face the internal network core, meaning they direct their CFM messages towards the central part of the MD rather than toward an external interface. This inward orientation allows Up MEPs to monitor connectivity within the internal layers of the network, effectively enabling operators to check the status of connections across multiple network segments. By sending messages towards the core, Up MEPs allow for comprehensive fault monitoring that extends across internal links and devices within the MD. This capability makes Up MEPs particularly useful in complex provider or operator networks where internal connectivity and seamless communication between network components need regular checks to ensure fault-free operations.

### 2.4.2 Down Mep

Conversely, **Down MEPs** are positioned to face outward, typically at the edge of a Maintenance Domain on an interface that connects to an external link. In this orientation, Down MEPs direct their CFM messages outward toward external networks or customer-facing links, making them well-suited for monitoring connections at the boundaries of an MD.

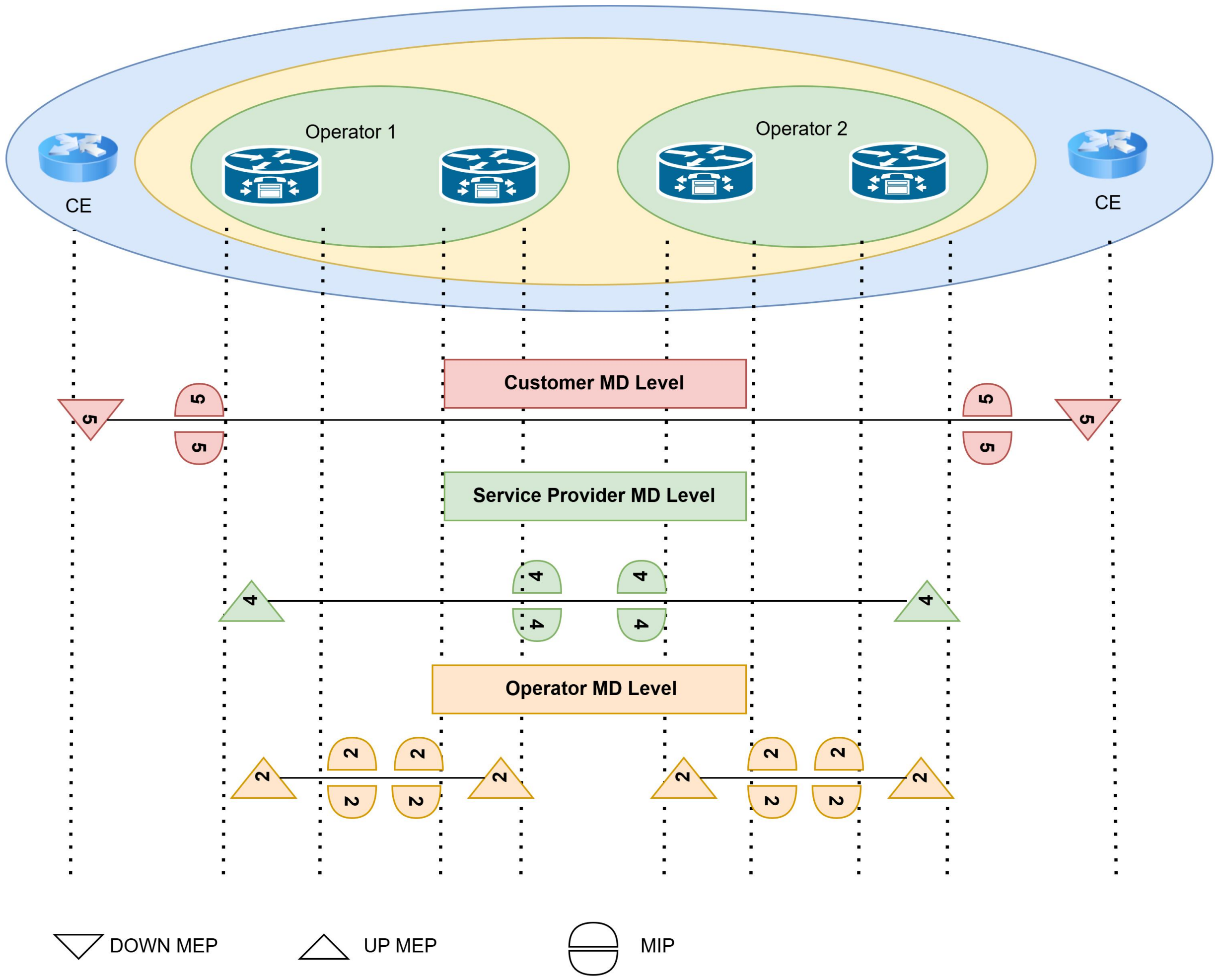
## 2.5 MIP

MIPs are located on a link between two MEPs within an MD, facilitating management. A higher number of MIPs present on a network increases the ease of network management and control. Carriers set up more MIPs for important services than they do for ordinary services.

# 3. Message Processing Rules in CFM

The following rules apply to CFM message processing-

1. If the MD level of the CFM message is higher than the MD level of the MEP/MIP, the MEP/MIP transparently passes the CFM message.
2. If the MD level of the CFM message is lower than the MD level of the MEP/MIP, the MEP/MIP discards the CFM message.
3. If the MD level of the CFM message is equal to the MD level of the MEP/MIP, the MEP/MIP processes the CFM message. Depending on the type of CFM message, the MEP/MIP responds to, transports to or accepts the message.




A MEP/MIP with a particular MD level generates a CFM message only at that level. In figure 2, the customer is interested in only the health of the end-to-end connection. Therefore, the MA has a high MD level (for example, MD=5). The MEPs/MIPs in provider and operator networks transparently pass customers CFM messages. The Provider is interested in the health of the service it provides. Therefore, the MA has a higher MD level (for example, MD=4) than the operators MD level but lower than the customer MD level. So, the MEPs/MIPs in operator network transparently pass the provider's CFM messages. The operators are interested only in the health of their own networks. Therefore, their MAs have lowest MD level (for example, MD=2). So, the CFM messages are confined to their own network.

# 4. Connectivity Fault Management protocols

There are three (3) protocols defined by CFM:

- Continuity Check Protocol
  - ✓ Fault Detection
  - ✓ Fault Notification
  - ✓ Fault Recovery

- Loopback Protocol
  - ✓ Fault Verification

- Linktrace Protocol
  - ✓ Path Discovery and Fault Isolation

## 4.1 Continuity Check Protocol

Continuity checks is another term for connectivity fault detection. This makes use of CCMs that are periodically sent by a MEP and received by RMEPs. CCMs can be used to detect connectivity faults across a link, or across a segment of a VLAN using VLAN aware MEPs, both of which are used within an MD/MA. CCMs are sent periodically to RMEPs at a given rate that is agreed upon by all the MEPs in the MA.

If an RMEP does not receive a CCM within a period three times the timeout interval at which CCMs are sent, the RMEP considers the path between itself and the MEP to be faulty.

### 4.1.1 Continuity Check (CC) Procedure

1. **CCM Generation**

- **Setup:** Each Maintenance Endpoint (MEP) is configured within a Maintenance Association (MA), meaning it shares a unique MAID and MD Level with other MEPs in the MA.
- **Periodic Transmission:** Each MEP is configured to periodically send a Continuity Check Message (CCM) using a multicast frame. The frequency is based on a timer set by the network operator, often configurable based on network requirements.
- **Sequence Numbers and RDI:** Every CCM includes a sequence number for tracking and a Remote Defect Indication (RDI) bit. The RDI bit indicates whether the transmitting MEP has received CCMs from all other MEPs in the MA.

2. **MEP Database Establishment**

- **CCM Reception Logging:** Each MEP maintains a MEP CCM Database. Here, it logs received CCMs indexed by the sending MEP's unique identifier (MEPID) to verify all configured MEPs are sending messages as expected.
- **MIP CCM Database (Optional):** Optionally, the Maintenance Intermediate Point (MIP) also logs received CCMs in an MIP CCM Database, cataloging each unique {FID, source_address, Port number} triple for tracking over time.

3. **Fault Identification**

- **Fault Conditions:** A MEP identifies faults when any of the following occur:
    - Failure to receive three consecutive CCMs from any configured MEP, which might signal a MEP or network failure.
    - CCMs received with incorrect MEPID, MAID, or MD Level indicate potential cross-connection or configuration errors.
    - The reception of a CCM containing a Port Status TLV or Interface Status TLV signals port-level faults like Bridge Port or interface failure.
- **Defect Priority:** Detected defects are prioritized. Cross-connect errors have the highest priority, followed by errors in CCM configuration, remote CCM reception, MAC status issues, and finally, RDI defects.

4. **CCM Termination**

- **CCM Timer Reset:** Each MEP has an internal timer that resets upon receipt of valid CCMs. If three CCMs are consecutively missed, the timer expires, and a defect is declared.
- **Fault Alarm Generation:** Once a defect is identified and confirmed, the MEP Fault Notification Generator issues a Fault Alarm, signaling the detected issue to the network administrator.
- **Alarm Resetting:** The Fault Alarm resets after a configurable period during which no further defects are indicated.

5. **CCM Session Monitoring and Maintenance**

- **RDI Monitoring:** MEPs check the RDI in received CCMs to detect unidirectional connectivity issues.
- **Sequence Number Checks:** Sequence numbers are used by receiving MEPs to identify and count any occasional losses in CCMs, allowing network administrators to preemptively address potential connectivity issues.

By following this procedure, the Continuity Check protocol enables reliable connectivity verification, fault detection, and reporting within the network, supporting ongoing fault management and quick troubleshooting.

## 4.1.2 CCM Packet Format

```
v Ethernet II, Src: Shanghai_09:0b:fd (ac:12:8e:09:0b:fd), Dst: OAM-Multicast-DA-Class-1_00
  > Destination: OAM-Multicast-DA-Class-1_00 (01:80:c2:00:00:30)
  > Source: Shanghai_09:0b:fd (ac:12:8e:09:0b:fd)
    Type: IEEE 802.1ag Connectivity Fault Management (CFM) protocol (0x8902)
v CFM EOAM 802.1ag/ITU Protocol, Type Continuity Check Message (CCM)
    000. .... = CFM MD Level: 0
    ...0 0000 = CFM Version: 0
    CFM OpCode: Continuity Check Message (CCM) (1)
v CFM CCM PDU
  v Flags: 0x05
      0... .... = RDI: 0
      .000 0... = Reserved: 0
      .... .101 = Interval Field: Trans Int 10s, max Lifetime 35s, min Lifetime 32.5s (5)
    First TLV Offset: 70
    Sequence Number: 487
    ...0 0000 0000 0001 = Maintenance Association End Point Identifier: 1
  v Maintenance Association Identifier (MEG ID)
      MD Name Format: Character String (4)
      MD Name Length: 4
      MD Name (String): test
      Short MA Name (MEG ID) Format: Character String (2)
      Short MA Name (MEG ID) Length: 2
      Short MA Name: aa
      Zero-Padding
  > Defined by ITU-T Y.1731
> CFM TLVs
```

| | |
|---|---|
| RDI | The Remote Defect Indicator bit. This is set by the sending MEP to indicate a defect has been detected. |
| CCM Interval | Specifies the agreed upon interval that all MEPs in the MA will use for sending CCMs. There are seven discrete values, ranging from 3.3 ms to 10 minutes. |
| Sequence Numbers | Can be 0 always, or incremented starting at 1. |
| MEP ID | A unique number that identifies the MEP and should be as unique as possible across the network. |
| MAID/MEGID | Carries the name of the MD and the MA for 802.1ag formats, or carries the name of the MA only for ITU Y.1731 formats. |
| TLVs | A Type Length Value (TLV). An optional field used to carry additional information. |

| | |
|---|---|
| | ✓ **Sender ID TLV** - the Bridge Identifier identifying which bridge the CCM originated from.<br>✓ **Port Status TLV** - the state of the Port (and VLAN) on which the sending MEP sits that originated the CCM message. The state is either blocked or forwarded.<br>✓ **Interface Status TLV** - the state of the interface on which the sending MEP sits that originated the CCM message. The state is either Up or Down or in Test. |

### 4.1.3 Connectivity defects

1. **RDI Defect (someRDIdefect)**

- Triggered when a remote MEP sends a CCM with the Remote Defect Indication (RDI) bit set, indicating an issue in its own connectivity.
- Cleared once all remote MEPs stop sending CCMs with the RDI bit set.

2. **Port Status Defect (someMACstatusDefect)**

- Declared when CCMs received from all remote MEPs report their bridge ports as "Blocked."
- Cleared if at least one remote MEP's CCM shows an unblocked port status.

3. **Interface Status Defect (someMACstatusDefect)**

- Triggered when one or more remote MEPs report their bridge interface status as "Down."
- Cleared when all remote MEPs report their interfaces as "Up."

4. **Missing CCM Defect (someRMEPCCMDefect)**

- Declared if a MEP fails to receive CCMs from one or more remote MEPs within 3.5 times the expected interval.
- Cleared when CCMs are consistently received within the expected interval from all remote MEPs.

5. **Error in CCM Configuration (errorCCMdefect)**

- Triggered if a received CCM has the correct MAID and Level but an incorrect or mismatched MEP ID or interval.
- Cleared once no configuration errors are detected over a monitoring period.

6. **Cross-Connection Defect (xconCCMdefect)**

- Occurs if a CCM arrives with the wrong MAID or MD Level, indicating a potential misconfiguration or cross-connect error.
- Cleared once no cross-connection errors are detected over a set period.

## 4.2 Loopback Protocol

Loopback is also called 802.1ag MAC ping. Similar to IP ping, loopback monitors connectivity of a path between a local MEP and an RMEP.

A MEP initiates an 802.1ag MAC ping test to monitor reachability of a MEP or MIP destination address. The source MEP and the monitored MEP or MIP must be of the same level and can be in the same MA or different MAs. The source MEP sends loopback messages (LBMs) to the MEP or MIP. After receiving these messages, the MEP or MIP replies with loopback reply messages (LBRs). Loopback helps locate faulty nodes, because faulty nodes cannot send LBRs in response to an LBM. LBMs and LBRs are unicast packets.

1. **LBM Generation**

- The Loopback Protocol starts with the transmission of a unicast Loopback Message (LBM) from a Maintenance Endpoint (MEP). This step is used for fault verification and isolation.
- The LBM includes specific parameters such as destination_address (MAC address of the target MEP or Maintenance Intermediate Point (MIP)), priority, and drop eligibility to control message handling.
- Each LBM contains a unique Loopback Transaction Identifier, which is incremented with each transmission, allowing the originating MEP to correlate Loopback Replies (LBRs) to specific LBMs.

```
˅ Ethernet II, Src: Shanghai_09:0b:fd (ac:12:8e:09:0b:fd), Dst: Shanghai_09:0b:f0 (ac:12:8e:09:0b:f0)
  › Destination: Shanghai_09:0b:f0 (ac:12:8e:09:0b:f0)
  › Source: Shanghai_09:0b:fd (ac:12:8e:09:0b:fd)
    Type: IEEE 802.1ag Connectivity Fault Management (CFM) protocol (0x8902)
˅ CFM EOAM 802.1ag/ITU Protocol, Type Loopback Message (LBM)
    000. .... = CFM MD Level: 0
    ...0 0000 = CFM Version: 0
    CFM OpCode: Loopback Message (LBM) (3)
˅ CFM LBM PDU
    Flags: 0x00
    First TLV Offset: 4
    Loopback Transaction Identifier: 2183843368
› CFM TLVs
```

2. **LBM Transmission**

- The LBM is sent from the originating MEP to the target MAC address, ensuring delivery to a particular MEP or MIP within the same Maintenance Association.
- While there's no set rate for sending LBMs, operators ensure the transmission rate does not overwhelm network queues.
- This step allows operators to confirm connectivity between specific network points under controlled conditions.

3. **LBM Reception and LBR Generation**

- Upon receiving the LBM, the target Maintenance Point (MP) checks the message for validity. Invalid LBMs (e.g., those with Group MAC addresses in source_address or mismatched destination_address) are discarded.
- If valid, the MP generates a Loopback Reply (LBR) based on the received LBM's data.
- The LBR mirrors the contents of the LBM but adjusts specific fields:
    - The LBM's source_address becomes the destination_address of the LBR.
    - The LBR's source_address is set to the responding MP's MAC address.
    - The OpCode field changes from LBM to LBR.
- The LBR is then sent back to the originating MEP.

```
˅ Ethernet II, Src: Shanghai_09:0b:f0 (ac:12:8e:09:0b:f0), Dst: Shanghai_09:0b:fd (ac:12:8e:09:0b:fd)
  > Destination: Shanghai_09:0b:fd (ac:12:8e:09:0b:fd)
  > Source: Shanghai_09:0b:f0 (ac:12:8e:09:0b:f0)
    Type: IEEE 802.1ag Connectivity Fault Management (CFM) protocol (0x8902)
˅ CFM EOAM 802.1ag/ITU Protocol, Type Loopback Reply (LBR)
    000. .... = CFM MD Level: 0
    ...0 0000 = CFM Version: 0
    CFM OpCode: Loopback Reply (LBR) (2)
˅ CFM LBR PDU
    Flags: 0x00
    First TLV Offset: 4
    Loopback Transaction Identifier: 2183843368
> CFM TLVs
```

4. **LBR Reception and Validation**

- When the originating MEP receives the LBR, it verifies that the destination_address matches its own MAC address to confirm it is the intended recipient.
- The MEP checks the Loopback Transaction Identifier in the LBR, comparing it to the identifier of recently sent LBMs. Based on whether the replies are in or out of sequence, the MEP updates respective counters.
- The MEP may also perform a bit-by-bit comparison between the LBR and the original LBM to identify any discrepancies, updating an error counter if differences are detected.

5. **Session Monitoring and Fault Analysis**

- The MEP continuously monitors LBM/LBR exchanges, using counters to detect patterns in packet loss, delays, or sequence errors.
- Discrepancies or out-of-sequence replies can indicate network congestion or potential hardware issues.
- By analyzing these metrics, administrators can assess connectivity quality, detect potential faults, and implement corrective measures if needed.

## 4.3 Linktrace Protocol

Linktrace is also called 802.1ag MAC trace. Similar to IP traceroute, it identifies a path between two MEPs.
A MEP initiates an 802.1ag MAC trace test to monitor reachability of a MEP or MIP destination address. The source MEP and the monitored MEP or MIP must be of the same level and can be in the same MA or different MAs. The source MEP constructs and sends a linktrace message (LTM) to the destination MEP or MIP. After receiving the LTM, each MIP forwards the LTM and replies with a linktrace reply message (LTR). Upon receiving the LTM, the destination MEP replies with an LTR message and does not forward the LTM. Based on the LTRs received, the source MEP obtains topology information about each hop on the path. LTMs are multicast packets and LTRs are unicast packets.

1. **Components Involved**

- **MEP Linktrace Initiator**: Responsible for initiating Linktrace Messages (LTMs).
- **Linktrace Responder**: Associated with a Bridge, processes LTMs and sends Linktrace Replies (LTRs).

2. **LTM Transmission**

- **Purpose**: An LTM is sent by the MEP Linktrace Initiator to discover the path to a target MAC address.
- **Frame Type**: Carried in a multicast frame with a destination address based on the MEP's Maintenance Domain (MD) Level.
- **Routing**: The LTM is relayed through the bridged network until it reaches a Maintenance Point (MP) at the appropriate MD Level.

```
˅ Ethernet II, Src: Shanghai_09:0b:fd (ac:12:8e:09:0b:fd), Dst: OAM-Multicast-DA-Class-2_00
  > Destination: OAM-Multicast-DA-Class-2_00 (01:80:c2:00:00:38)
  > Source: Shanghai_09:0b:fd (ac:12:8e:09:0b:fd)
    Type: IEEE 802.1ag Connectivity Fault Management (CFM) protocol (0x8902)
˅ CFM EOAM 802.1ag/ITU Protocol, Type Linktrace Message (LTM)
    000. .... = CFM MD Level: 0
    ...0 0000 = CFM Version: 0
    CFM OpCode: Linktrace Message (LTM) (5)
˅ CFM LTM PDU
  ˅ Flags: 0x80
      1... .... = UseFDBonly: 1
      .000 0000 = Reserved: 0
    First TLV Offset: 17
    Linktrace Transaction Identifier: 2181523496
    Linktrace TTL: 64
    Linktrace Message: Original Address: Shanghai_09:0b:fd (ac:12:8e:09:0b:fd)
    Linktrace Message:   Target Address: Shanghai_09:0b:f0 (ac:12:8e:09:0b:f0)
> CFM TLVs
```

3. **LTM Processing**

- The MP intercepts the LTM and forwards it to the Bridge's Linktrace Responder.
- The Linktrace Responder determines if the target MAC address can be forwarded to a single egress Bridge Port.
- If the target MAC is valid, it sends a unicast LTR back to the originating MEP after a random delay (0 to 1 second) to avoid congestion.

4. **Linktrace Reply (LTR) Generation**

- The LTR contains information regarding the path taken by the LTM.
- If the LTM was received by a Maintenance Hop Function (MHF), the Linktrace Responder also forwards a modified version of the LTM towards the target MAC address.

```
⌄ Ethernet II, Src: Shanghai_09:0b:f0 (ac:12:8e:09:0b:f0), Dst: Shanghai_09:0b:fd
  > Destination: Shanghai_09:0b:fd (ac:12:8e:09:0b:fd)
  > Source: Shanghai_09:0b:f0 (ac:12:8e:09:0b:f0)
    Type: IEEE 802.1ag Connectivity Fault Management (CFM) protocol (0x8902)
⌄ CFM EOAM 802.1ag/ITU Protocol, Type Linktrace Reply (LTR)
    000. .... = CFM MD Level: 0
    ...0 0000 = CFM Version: 0
    CFM OpCode: Linktrace Reply (LTR) (4)
⌄ CFM LTR PDU
  ⌄ Flags: 0xa0
      1... .... = UseFDBonly: 1
      .0.. .... = FwdYes: 0
      ..1. .... = TerminalMEP: 1
      ...0 0000 = Reserved: 0
    First TLV Offset: 6
    Linktrace Transaction Identifier: 2181523496
    Linktrace TTL: 63
    Linktrace Reply Relay Action: RlyHit (1)
> CFM TLVs
```

### 5. **Path Discovery Challenges**

- In connectionless environments like Bridged LANs, MAC addresses may be deleted from the Filtering Database after topology changes.
- To address the aging of MAC addresses, CFM (Connectivity Fault Management) provides three strategies:
  - Automatically launch Linktrace protocols post-fault detection.
  - Maintain destination MP information at intermediate points.
  - Issue periodic LTMs during normal operations.

### 6. **LTM Origination**

- LTMs are transmitted in response to operator commands.
- Each LTM retains a unique Transaction Identifier (TID) for at least 5 seconds to correlate with returned LTRs.
- LTMs are indexed in a Linktrace database for tracking.

### 7. **Reception and Forwarding of LTMs**

- LTMs are forwarded as multicast frames until they reach an appropriate MP.
- Invalid LTMs are discarded without processing.
- The Linktrace Responder processes the LTM, checks the TTL, and sends an LTR if conditions are met.

8. **Conditions for LTR Responses**

- An LTR is sent if:
  - An ordinary data frame could be forwarded through exactly one egress port.
  - The Target MAC address is found in the Bridge's MIP CCM Database.
  - The MAC address of the MP receiving the LTM matches the Target MAC address.
- The LTM TTL must be greater than zero.

9. **LTR Reception**

- LTRs are validated upon receipt by the MEP Linktrace Initiator.
- Invalid LTRs are discarded.
- Valid LTRs create new entries in the Linktrace database for record-keeping.

10. **Future Considerations**

- The protocol accommodates potential future extensions by forwarding unknown TLVs in both LTMs and LTRs, allowing for new capabilities without breaking existing functionality.

# 5. Conclusion

This report detailed the key aspects of Connectivity Fault Management (CFM) in bridged networks, emphasizing Maintenance End Points (MEPs) and protocols that support connectivity and troubleshooting. Key protocols—Continuity Check, Loopback, and Linktrace—work together to monitor network paths, validate segments, and trace routes to specific MAC addresses, enabling rapid fault detection and isolation.

CFM's state machines, timers, and defect notifications enhance network reliability and allow for efficient management of faults. Its extensibility for future updates further supports evolving network needs, making CFM a resilient solution for maintaining network stability.